

Dual Porosity / Dual Permeability Model

The following describes the governing equations for the dual-porosity and dual-permeability approach to modelling naturally fractured reservoirs. The governing equations are an extension of the equations for single-porosity systems described in [Nghiem and Li \(1988\)](#). The representation of the matrix follows [Kazemi et al \(1976\)](#), where fractures are assumed orthogonal in the three directions and acts as boundaries to matrix elements.

Dual Porosity Formulation - Matrix Blocks

$$\psi_{im} \equiv -\tau_{iomf} - \tau_{igmf} - \frac{V}{\Delta t} (N_i^{n+1} - N_i^n)_m = 0 \quad i = 1, \dots, n_c \quad (\text{A6.1})$$

$$\psi_{n_c+1,m} \equiv -\tau_{wmf} - \frac{V}{\Delta t} (N_{n_c+1}^{n+1} - N_{n_c+1}^n)_m = 0 \quad (\text{A6.2})$$

$$\psi_{pm} \equiv \sum_{i=1}^{n_c+1} N_{im}^{n+1} - \phi_m^{n+1} (\rho_o S_o + \rho_g S_g + \rho_w S_w)_m^{n+1} = 0 \quad (\text{A6.3})$$

Dual Porosity Formulation - Fracture Blocks

$$\begin{aligned} \psi_{if} \equiv & \Delta T_{of}^s y_{iof}^s (\Delta p^{n+1} - \gamma_o^s \Delta D)_f + \Delta T_{gf}^s y_{igf}^s (\Delta p^{n+1} + \Delta p_{cog}^s - \gamma_g^s \Delta D)_f \\ & + q_i^{n+1} + \tau_{iomf} + \tau_{igmf} - \frac{V}{\Delta t} (N_i^{n+1} - N_i^n)_f = 0 \quad i = 1, \dots, n_c \end{aligned} \quad (\text{A6.4})$$

$$\psi_{n_c+1,f} \equiv \Delta T_{wf}^s (\Delta p^{n+1} - \Delta p_{cwo}^s - \gamma_w^s \Delta D)_f \quad (\text{A6.5})$$

$$+q_w^{n+1} + \tau_{wmf} - \frac{V}{\Delta t} (N_{n_c+1}^{n+1} - N_{n_c+1}^n)_f = 0$$

$$\psi_{pf} \equiv \sum_{i=1}^{n_c+1} N_{if}^{n+1} - \phi_f^{n+1} (\rho_o S_o + \rho_g S_g + \rho_w S_w)_f^{n+1} = 0 \quad (\text{A6.6})$$

where N_i , $i = 1, \dots, n_c + 1$ are the moles of i per unit of gridblock volume. The subscript i with $i = 1, \dots, n_c$ corresponds to the hydrocarbon components and the subscript $n_c + 1$ denotes the water component. τ_{iomf} and τ_{igmf} are the matrix-fracture transfer in the oil and gas phases for component i , and τ_{wmf} is the matrix-fracture transfer for water. The superscripts n and $n + 1$ denote respectively the old and current time level and the superscript s refers to n for explicit blocks and to $n + 1$ for implicit blocks. The subscript f and m correspond to the fracture and matrix respectively.

Dual-Permeability Formulation

This formulation is similar to the dual-porosity formulation, except that matrix blocks are connected to one another and thus provide alternate channels for fluid flow. The fracture equations are the same as those in dual-porosity formulation. The matrix flow equations contain additional terms as follows:

$$\psi_{im} = \Delta T_{om}^s y_{iom}^s (\Delta p^{n+1} - \gamma_o^s \Delta D)_m + \Delta T_{gm}^s y_{igm}^s (\Delta p^{n+1} + \Delta p_{cog}^s - \gamma_g^s \Delta D)_m \quad (\text{A6.7})$$

$$-\tau_{iomf} - \tau_{igmf} - \frac{V}{\Delta t} (N_i^{n+1} - N_i^n)_m = 0 \quad i = 1, \dots, n_c$$

$$\psi_{n_c+1,m} = \Delta T_{wm}^s (\Delta p^{n+1} - \Delta p_{cwo}^s - \gamma_w^s \Delta D)_m$$

$$-\tau_{wmf} - \frac{V}{\Delta t} (N_{n_c+1}^{n+1} - N_{n_c+1}^n)_m = 0 \quad (\text{A6.8})$$

Matrix - Fracture Transfer, NTRAN = 0

Following [Kazemi et al \(1976\)](#):

$$\tau_{jmf} = \sigma V \frac{k_{rj} \rho_j}{\mu_j} (p_{jm} - p_{jf}) \quad j = o, g, w \quad (\text{A6.9})$$

where

$$p_g = p_o + p_{cog}$$

$$p_w = p_o + p_{cwo}$$

$$\sigma = 4 \left(\frac{kx}{l_x^2} + \frac{ky}{l_y^2} + \frac{kz}{l_z^2} \right)$$

with V being the gridblock volume and I_x , I_y and I_z the fracture spacings in the x , y and z directions respectively. The matrix and fracture blocks are assumed to be at the same depth, no gravity term is included in Equation (A6.9) explicitly.

Generally, the matrix capillary pressure is much larger than fracture capillary pressures and Equation (A6.9) cannot model correctly the gravity drainage processes.

Matrix - Fracture Transfer, NTRAN = 1

The gravity effects can be included by assuming complete gravity segregation of the oil, gas and water phases as follows:

$$\tau_{omf} = \sigma V \frac{k_{ro} \rho_o}{\mu_o} (p_{om} - p_{of}) \quad (\text{A6.10})$$

$$\begin{aligned} \tau_{gmf} = \sigma V \frac{k_{rg} \rho_g}{\mu_g} & \left[p_{gm} - p_{gf} + \left(\frac{S_g}{1 - S_{org} - S_{wr}} - \frac{1}{2} \right)_m \Delta\gamma_{ogm} h \right. \\ & \left. - \left(\frac{S_g}{1 - S_{org} - S_{wr}} - \frac{1}{2} \right)_f \Delta\gamma_{ogf} h \right] \end{aligned} \quad (\text{A6.11})$$

$$\tau_{wmf} = \sigma V \frac{k_{rw} \rho_w}{\mu_w} \left[p_{wm} - p_{wf} - \left(\frac{1}{2} - \frac{S_w - S_{wr}}{1 - S_{orw} - S_{wr}} \right)_m \Delta\gamma_{wom} h \right. \\ \left. + \left(\frac{1}{2} - \frac{S_w - S_{wr}}{1 - S_{orw} - S_{wr}} \right)_f \Delta\gamma_{wof} h \right] \quad (\text{A6.12})$$

where

- $\Delta\gamma_{og}$ = (oil mass density - gas mass density) g
- $\Delta\gamma_{wo}$ = (water mass density - oil mass density) g
- h = height of matrix element in the direction of gravity and subscripts
- f = fracture property
- m = matrix property

Note that normally $\Delta\gamma_m = \Delta\gamma_f$ and the additional gravity terms just accounts for the differences in the height of the fluid columns. Evaluations of Equations [\(A6.10\)](#) through [\(A6.12\)](#) are simple and very little additional work is required.

Matrix - Fracture Transfer, NTRAN = 2

In deriving Equations [\(A6.10\)](#) through [\(A6.12\)](#), the effects of capillary pressures and gravity are considered separately and later summed together. A more rigorous approach is to use the pseudo capillary pressures. Assuming instantaneous vertical gravity - capillary equilibrium, pseudo capillary pressures can be derived as follows:

$$\int_{\tilde{p}_{cog} - \gamma_{og} h/2}^{\tilde{p}_{cog} + \gamma_{og} h/2} S_g dp_{cog} - \gamma_{og} h S_g = 0 \quad (\text{A6.13})$$

$$\int_{\tilde{p}_{cwo}-\gamma_{wo}h/2}^{\tilde{p}_{cwo}+\gamma_{wo}h/2} S_w dp_{cwo} - \gamma_{wo} h S_w = 0 \quad (\text{A6.14})$$

where $\tilde{p}_{cog}$ and $\tilde{p}_{cwo}$ are the pseudo p_{cog} and pseudo p_{cwo} respectively which accounts for both the capillary and gravity effects.

$$\tau_{omf} = \sigma V \frac{k_{ro} \rho_o}{\mu_o} (p_{om} - p_{of}) \quad (\text{A6.15})$$

$$\tau_{gmf} = \sigma V \frac{k_{rg} \rho_g}{\mu_g} (p_{om} + \tilde{p}_{cog,m} - p_{of} - \tilde{p}_{cog,f}) \quad (\text{A6.16})$$

$$\tau_{wmf} = \sigma V \frac{k_{rw} \rho_w}{\mu_w} (p_{om} - \tilde{p}_{cwo,m} - p_{of} + \tilde{p}_{cwo,f}) \quad (\text{A6.17})$$

The pseudo capillary pressures are functions of fluid saturations, densities as well as shape of rock capillary pressure curves. They change dynamically with the change in fluid saturations and densities. As these pseudo capillary pressures appear in the limit of integrals, they are evaluated iteratively and can be time-consuming. Generally, Equations [\(A6.13\) through \(A6.17\)](#) gives results similar to those in Equations [\(A6.10\) through \(A6.12\)](#). In fact, the latter could be derived from the former by assuming negligible rock capillary pressures. Equations [\(A6.13\) through \(A6.17\)](#) will be used when NTRAN = 2 in CARD RES4.

Matrix - Fracture Transfer, NTRAN = 3

All earlier equations assumes that transfer of all phases occurs through the entire surfaces of the matrix element. This is not true when the matrix element is partially immersed in either gas or water. To account for this effect, Equations [\(A6.15\) through \(A6.17\)](#) were modified as follows:

$$\tau_{omf} = \sigma V \frac{k_{ro} \rho_o}{\mu_o} (p_{om} - p_{of}) \quad (\text{A6.18})$$

$$\tau_{gmf} = \sigma V \frac{k_{rg} \rho_g}{\mu_g} \left\{ (p_{om} - p_{of}) + \left[S_{gm} + \frac{\sigma_z}{\sigma} \left(\frac{1}{2} - S_{gm} \right) \right] (\tilde{p}_{cog,m} - \tilde{p}_{cog,f}) \right\} \quad (\text{A6.19})$$

$$\begin{aligned} \tau_{wmf} = \sigma V \frac{k_{rw} \rho_w}{\mu_w} \{ & (p_{om} - p_{of}) - (p_{cwo,m} - p_{cwo,f}) \\ & - \left(\frac{1}{2} \frac{\sigma_z}{\sigma} \right) [(\tilde{p}_{cwo,m} - \tilde{p}_{cwo,f}) - (p_{cwo,m} - p_{cwo,f})] \} \end{aligned} \quad (\text{A6.20})$$

The form of these equations was found to give reasonable results. Similarly, the evaluations of the pseudo capillary pressures requires iterations and can be time-consuming.